MySQL Data Exploration & Data Cleaning

A practical SQL project exploring grocery product inventory, checking data quality and analysing pricing, discounts and stock.

123
records*
Derived from stock counts

10
columns
Displayed CREATE TABLE

2
categories
Observed in result outputs

Dataset & table structure

Database
zepto

Main queried table
zepto.zepto_v2

Columns in displayed CREATE TABLE statement

- `sku_id`
- `name`
- `discountpercent`
- `discountedsellingprice`
- `outofstock`
- `category`
- `mrp`
- `availablequantity`
- `weightingram`
- `quantity`

Note: CREATE TABLE names `zepto`, while analysis queries use `zepto.zepto_v2`. The screenshots do not prove these are the same table.

Project workflow

01 **Explore structure**
02 **Check NULLs & duplicates**
03 **Inspect stock status**
04 **Review price conversion**
05 **Analyse discounts & value**
06 **Classify product weights**

*123 = 117 in stock + 6 out of stock; raw COUNT(*) output not visible.

Data exploration

SQL checks used to inspect records, missing values, categories, stock availability and repeated product names.

Core exploration queries

```
SELECT COUNT(*) FROM zepto.zepto_v2;
SELECT * FROM zepto.zepto_v2 LIMIT 10;
SELECT DISTINCT category
FROM zepto.zepto_v2 ORDER BY category;
```

NULL-value check

0 matching rows

Checked name, category, mrp, discountpercent, availablequantity, discountedSellingPrice, weightInGms, outofstock and quantity.

Stock availability

117 In stock

6 Out of stock

Repeated product names

```
SELECT name, COUNT(*)
FROM zepto.zepto_v2
GROUP BY name
HAVING COUNT(*) > 1;
```

6 names shown • 2 SKUs each

Onion	Potato
Beetroot	Carrot
Arden Eggs White	Amul Pure Ghee (Pouch)

A second query grouped by category and name and returned the same six repeated name/category combinations, each with duplicate_count = 2.

Data cleaning & transformation

The screenshots show checks and SQL statements intended to identify invalid prices and convert stored paise values into rupees.

1. Identify zero-price records

```
SELECT *
FROM zepto.zepto_v2
WHERE mrp = 0
  AND discountedSellingPrice = 0;
```

Purpose: locate products where both MRP and discounted selling price are zero. The matching result is not visible, so the number of affected rows is unknown.

2. Convert paise to rupees

```
SET SQL_SAFE_UPDATES = 0;
UPDATE zepto.zepto_v2
SET mrp = mrp / 100.0,
    discountedSellingPrice =
    discountedSellingPrice / 100.0;
SET SQL_SAFE_UPDATES = 1;
```

The conversion divides both price columns by 100. The UPDATE statement is visible in the editor, but a clear successful UPDATE response is not shown.

Validation notes

- Price values are subsequently used in price-per-gram and revenue queries.
- No before/after comparison or final cleaned-row count is visible.
- The conversion statement appears twice in the editor; confirm only one conversion was applied.

Business analysis & key findings

Additional queries explored discount performance, estimated inventory revenue, price per gram and product weight bands.

Estimated revenue by category

Cooking Essentials

45,033

Fruits & Vegetables

10,846

SUM(discountedSellingPrice * availablequantity), grouped by category.

Average discount

Fruits & Vegetables

15.46%

Cooking Essentials

5.13%

Grouped by category and sorted by average discount descending.

Price per gram • 10 rows displayed

```
ROUND(discountedSellingPrice
 / weightInGms, 2)
AS price_per_gram
WHERE weightInGms > 108
```

Observed range

0.02 – 0.04

Displayed examples include Onion, Tata Salt, Beetroot, Carrot and Potato products.

Weight classification

LOW

≤ 100 g

MEDIUM

101–500 g

BULK

> 500 g

Implemented with CASE WHEN weightInGms thresholds. Sample rows shown are in the Low band; category totals are not shown.

Tools & learning outcomes

MySQL • SQL • GROUP BY • HAVING • CASE • Aggregations

Data exploration • Data quality checks • Pricing analysis • Inventory insights

Revenue output currency is not explicitly labelled in the result grid.